

with Light-Up Circuits

Course Overview

- **Grade level:** 2nd
- **Duration:** 90 minutes
- **Class size:** Up to 24 students
- **Location:** Steamspace @ Ft. Apache

Timeline

T i m e	Activity	S e c t i o n
0 : 0 0 - 0 : 1 0	Introduction & Safety briefing	S e c t i o n 1
0 : 1 0 - 0 : 2 0	Choose animal, trace & cut cardstock	S e c t i o n 3
0 : 2 0	Punch holes & build with LEGO bricks	S e c t i

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:		n
5		
0		4
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0		t
-	Add LED circuit (optional)	i
0		o
:		n
7		
5		5
0		S
:		e
7		c
5		t
-	Reflect & share	i
0		o
:		n
8		
5		6
0		S
:		e
8		c
5		t
-	Cleanup	i
0		o
:		n
9		
0		6

Materials List (per class of 24)

Required — Paper & LEGO

Item	Quantity	Notes
Cardstock paper	50 sheets	8.5×11", various colors

(assorted colors)

Animal template printouts

30 (6 per animal × 4 animals + extras)

Print on regular paper for tracing

Hole punch stencils

8–12 sets

From Brown Dog Gadgets — printable PDF or 3D printed

Hole punch tools

8–12

Standard single-hole punch

Scissors

24

Child-safe

LEGO bricks (assorted)

500+ pieces

Mixed colors, various sizes (1×1, 1×2, 2×2 studs)

3D-printed special shapes

2 sets

Bear ears, eagle wing tips, fish fins, butterfly antennae (STL files from BDG)

Optional — LED Circuit (per student who opts in)

Item	Quantity	Notes
CR2032 coin battery	25	1 per student + extras

LED (assorted colors)	2 5	3mm or 5mm, mixed colors
	2	
Maker Tape (copper tape)	r o l l s	Brown Dog Gadgets Maker Tape or equivalent conductive tape
Small cardstock squares	2 5	~2×2" pieces for circuit backing

Tools & Supplies

Item	Quantity	Notes
Pencils	2 4	For tracing templates
Paperclips	5 0	To hold stencils in place while punching
	4	
Tape (regular)	r o l l s	For attaching circuit backing to animal
Table	6	Optional — protect work

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For photos of finished
projects

Pre-Class Preparation

Before the Field Trip (1–2 hours prep)

1. **Print animal templates** — Print 6 copies of each animal outline (bear, eagle, fish, butterfly) on regular paper. These are for tracing onto cardstock.

1. **Print/prepare hole punch stencils** — Download from [Brown Dog Gadgets LEGO Hole Punch Stencils](#). Options:

- Print on heavy cardstock and cut out the hole marks
- 3D print the stencil files (more durable, reusable)
- Laser cut from acrylic (most durable)

1. **3D print special pieces** (if you have a 3D printer):

- Bear ears (2 per bear)
- Eagle wing tips (2 per eagle)
- Fish fins (2–3 per fish)
- Butterfly antennae (2 per butterfly)
- Use the STL files from Brown Dog Gadgets or design your own
- Print in PLA, 0.2mm layer height, ~20% infill
- Print ahead of time — these take 15–30 min each

1. **Sort LEGO bricks** — Separate into color bins for easy student

access. Include 1×1, 1×2, and 2×2 stud bricks primarily.

1. Prep LED circuit kits (if offering the LED option):

- Pre-cut Maker Tape into ~3" strips (4 strips per student)
- Pre-cut small cardstock squares (~2×2")
- Test LEDs and batteries — discard any dead ones
- Consider pre-bending LED legs for younger students

1. Set up tables — 4–6 tables with 4 students each. Each table needs:

- Scissors, pencils, hole punch
- Stack of cardstock
- Bin of LEGO bricks
- Animal templates
- Stencil set

During the Field Trip

- **Circulate during building** — help students who are struggling with hole punching or snapping bricks
- **LED station** — Set up a separate table for LED circuit assembly with extra supervision
- **Take photos** throughout — capture the building process and finished projects
- **Watch the clock** — keep sections moving so students finish within 90 minutes

Post-Field Trip

- Upload photos to the Moodle course (Section 6 — Reflect & Share)
- Have students fill out the reflection form

- Collect and store reusable materials (stencils, 3D-printed pieces, leftover bricks)

Source Links

- [Brown Dog Gadgets — LEGO Hole Punch Stencils](#)
- [Brown Dog Gadgets — Getting Started with Paper Circuits](#)
- [Brown Dog Gadgets Shop](#) — for kits, Maker Tape, LEDs, batteries
- [Arizona Science Standards](#)
- [NGSS K-2 Engineering Design](#)

Safety Reminders

- Coin batteries are a choking/swallowing hazard — supervise closely, count batteries in and out
- Scissors safety — cut away from body, walk with points down
- LED legs can be sharp — bend them for students if needed
- 3D-printed pieces can break — don't allow bending or twisting